

QQ Super Compression

User Manual / 1.2.6

Control dynamics. Keep the attack natural.

For vocals, instruments, buses and mastering. There are no conventional Attack / Release controls: shape dynamics with Lookahead, Ratio, Threshold and Mix.

Classic uses a fixed dB ratio; Super retains the original curve. Both algorithms support Single processing, upward boost above a gate, and Dual with separate upward and downward controls.



Start with what you hear

Decide how steady the dynamics should be and how much articulation to retain. The Display helps explain changes; it does not replace listening.

Quick start and navigation

A first session in six steps.

- 1 Insert on the track to process**
 Start with the CLASSIC algorithm, SINGLE, ST and SC: INT so the plugin follows the track itself.
- 2 Choose Lookahead**
 26 ms is a practical starting point, not a rule for every source.
- 3 Start at 1:1**
 All Ratios default to 1:1, with no dynamic change. To try downward processing, raise Ratio and leave Range OFF. For upward and Dual processing, see pages 8-11.
- 4 Use Mix to set the processing amount**
 If processing is too strong, lower Mix to blend back the original dynamics. Both boost and cut displays follow Mix.
- 5 Compensate level and compare**
 Play a representative passage, use Match, then trim Makeup / Output Gain to avoid loudness bias.
- 6 Listen in the full mix**
 Use Bypass or A/B to judge articulation, consistency and groove - not only the GR number.

Find a topic	Pages
Interface, themes and gestures	3-4
Classic / Super, Single and upward processing	5-8
Dual, branch switches and Ratio LINK	9-11
Input / Output LINK, Match and A/B	12-14
Display and Lookahead	15-17
ST / LR / MS and channel LINK	18-20
Sidechain panel, internal HPF and external routing	21-23
Mixing examples, parameter reference and help	24-28

Know the interface

The same audio controls remain in place across all themes.



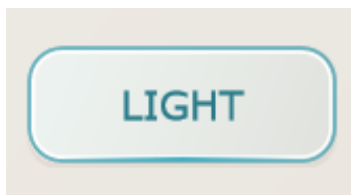
Light / ST. The values illustrate the interface; they are not a recommended preset.

Area	Use it to
Display	View historical level and gain references under current settings; see page 15.
Two boundary faders	Threshold / Range in Single; UP / DOWN thresholds in Dual.
Input / Output / Gain +/-	Check input, output and effective dynamic gain.
Bottom knobs and SINGLE / DUAL	Adjust gain, Ratio and Mix; switch Single / Dual beside Ratio.
Mode / LINK / Monitor	Choose channel processing, link controls or audition domains.
Top right	Switch Classic / Super, open sidechain, change theme, use A/B and Bypass.

The interface uses a 3:2 landscape aspect ratio. Drag the bottom-right corner to resize it. The plugin remembers the last size when reopened.

Themes and everyday controls

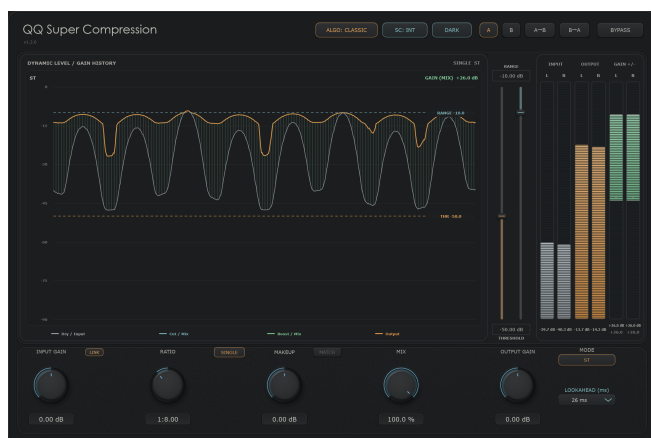
Click the top-right button labelled LIGHT, DARK or CLASSIC to cycle themes. The last selection is restored when the plug-in opens again.



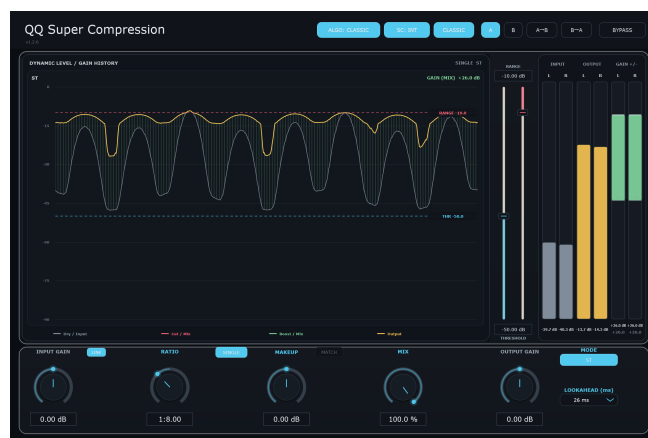
The theme button is between SC and A/B.

Theme	Appearance
Light	Light interface.
Dark	Subdued dark interface.
Classic	Original classic dark interface.

Themes change appearance only. The CLASSIC theme is separate from ALGO: CLASSIC: one changes the visual style, the other changes the processing algorithm.



Dark

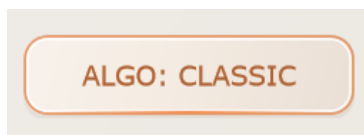


Classic

Gesture	Result
Drag / Shift-drag	Normal / fine adjustment.
Double-click a value	Type an exact value.
Alt + left click	Restore the default. When linked, related controls follow within their allowed ranges.
Ctrl/Cmd + Z	Undo; add Shift to redo.

Classic and Super: two dynamic curves

Switch with ALGO at the top. Both algorithms work in SINGLE and DUAL.



Algorithm	What Ratio means	How to approach it
Classic	A fixed dB ratio above the threshold.	Use familiar threshold and ratio values to decide how much of the excess level remains.
Super	The original QQ dynamic curve, not a fixed dB slope.	Choose by processing strength, detail and the sound after matching loudness.

Classic downward: compress the excess above threshold

Set Threshold to -10 dB, Ratio to 5:1 and Range to OFF. A steady detector level of 0 dB is 10 dB above the threshold. That excess becomes 2 dB, giving -8 dB after processing. A detector level of -12 dB stays below the threshold and receives no dynamic attenuation.

Conditions for this example

Use internal detection, 0 dB Input, Makeup and Output Gain, and 100% Mix. This describes the curve for a steady detector level, not a hard limit on individual audio samples. For changing signals and the Display, see pages 15-16.

Upward processing still boosts above the lower gate

In both algorithms, the lower threshold enables boost above it. Classic uses the dB slope corresponding to the reciprocal Down Ratio; Super uses the original upward curve. Both share the same active intervals, Range behavior and Dual handoff rules.

Minimum threshold and remembered choice

Classic stops at -90 dB and continues using the Classic curve at the bottom. Super retains -inf, meaning no lower gate. New instances remember the last algorithm; projects and A/B recall their saved choice. A short crossfade reduces switching transients.

Ratio, Input and Mix

Three useful controls, not interchangeable settings.



Ratio

Single centers on 1:1: above it, processing is downward; below it, processing is upward. Within the active interval, larger downward values generally mean more cut, and smaller upward values mean more boost. Dual provides separate UP RATIO and DOWN RATIO controls; see page 9.

See page 5 for the Classic / Super distinction. Single ranges from 1:1000 to 1000:1, and all Ratios default to 1:1. A very high value does not turn the plugin into a brick-wall limiter.

Input Gain

Input Gain changes the level entering the processing path, including the Dry path. With INT it also changes the detector level relative to Threshold. With EXT, the external send and Key Gain set the detector level; Input Gain is not a substitute.

Mix

Mix blends Dry with processed Wet. It is also a useful processing amount control: first make the boost or cut clear, then lower Mix to blend back the original dynamics you want to retain.

Dynamic gain includes Mix

At Mix = 0%, effective dynamic gain is 0 dB; at 100%, the full processing amount is retained. Intermediate values are not simply the gain in dB multiplied by Mix. Makeup and Output Gain are excluded from this dynamic gain.

A practical order: find the compression character with Ratio, set its final strength with Mix, adjust Input Gain when you need a different input level, then compare loudness again.

Single: Threshold and Range

Choose the active levels, then set direction and strength with Ratio.



SINGLE: the lower Threshold and upper Range define the processing interval. The faders align with the dashed lines in the display.

Detector level	Dynamic processing
At or below Threshold	No dynamic boost or attenuation.
Above Threshold and below Range	Upward or downward processing, according to Ratio.
At or above a finite Range	Returns fully to 0 dB dynamic gain.

Range OFF: no upper cutoff

At its top OFF position, Range imposes no upper limit on the selected algorithm. Pull it down to set a cutoff. A finite 0 dB setting differs from OFF. Classic Threshold stops at -90 dB; Super retains -inf, meaning no lower gate.

Near a finite Range, dynamic gain returns continuously to 0 dB from inside the active interval, avoiding a sudden jump across the boundary. This transition follows detector level; it adds no Attack time or extra latency.

When the faders meet

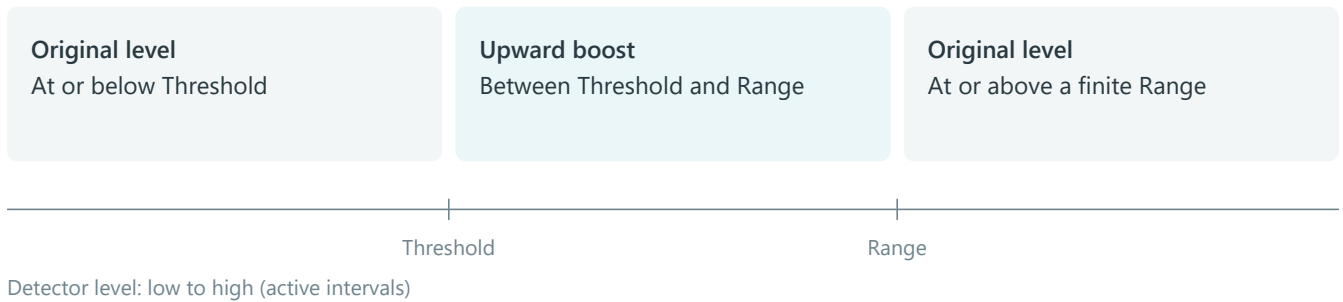
Range stays at or above Threshold. When one fader reaches the other, it moves the other along; coincident boundaries disable dynamic processing. Drag back to separate them.

Here, 0 dB dynamic gain means no dynamic boost or cut. Input, Makeup, Mix and Output Gain still follow their settings. In EXT, these boundaries apply to the external detector signal.

Upward: boost above the gate

Classic and Super both use this gate rule to bring selected detail forward.

In SINGLE, move Ratio below 1:1, for example to 1:2, 1:4 or 1:8, to enter upward processing. The displayed 1:8 means 1/8. Smaller values generally give eligible material more boost.



The gate enables boost; it does not mute audio

Threshold acts as a gate that allows boost above it. Below the gate, the original level is retained: quieter sounds are neither muted nor specifically boosted.

1 Select the detail to bring forward

Select SINGLE. Set a lower Threshold and upper Range, for example -50 dB / -10 dB. Listen for the detail you want between them. These values are examples only.

2 Then adjust Ratio and Mix

Move gradually from 1:1 toward 1:2 and 1:4. If the boost is too strong, reduce its strength or lower Mix. Do not substitute Makeup for dynamic boost.

3 Listen to quiet and loud passages

Boost enters smoothly near the lower gate. Beyond that transition, quieter parts in the active interval generally receive more boost. Boost approaches 0 dB near the upper boundary. Range OFF removes the finite upper cutoff.

To leave very quiet breaths, noise or reverb at their original level, place the lower gate above them. Check that the desired material still enters the boost interval. Selection follows detector level, not sound type.

Dual: lift quieter parts, control louder ones

Click the small button beside Ratio to change SINGLE to DUAL.



DUAL: UP RATIO and DOWN RATIO control the two directions separately. UP / DOWN thresholds replace Threshold / Range.



Detector level: low to high (active intervals)

At DOWN, dynamic gain returns to 0 dB. Above it, upward processing stops and only Down Ratio applies. The two gains never stack at the same detector level. Dual has no Range control.

Starting a new instance

UP: -90 dB in Classic, -inf in Super. DOWN: 0 dB.

How to read it

The thresholds start at opposite ends; this does not disable upward processing.

Up Ratio = 1:1; Down Ratio = 1:1

Both directions are neutral. Adjust a Ratio to hear processing.

Threshold order and collision

DOWN must stay at or above UP. When one fader reaches the other, it pushes the other along. Dynamic processing stops when the thresholds coincide.

Dual switches: audition each direction

A small ON / OFF switch sits beside each UP RATIO and DOWN RATIO knob.



Both branches on. Each switch preserves its Ratio value for repeated comparison.

UP / DOWN	Processing you hear
ON / ON	Boost between UP and DOWN; attenuation above DOWN.
ON / OFF	Upward only. Boost still stops at DOWN.
OFF / ON	Downward only. No dynamic processing below DOWN.
OFF / OFF	No dynamic boost or attenuation. Other gain and Mix settings still apply.

- Set up the full Dual sound**
 Set both thresholds and Ratios. Play material with quiet and loud passages.
- Hear what each direction contributes**
 Switch DOWN off to hear Up, or UP off to hear Down, then enable both. There is no need to return to SINGLE or reset either Ratio.
- Check the display and compare at similar loudness**
 The switches update dynamic gain and the Display history reference. Compensate overall level, then compare detail, articulation and consistency.

Crossfaded switching; boundaries still apply

Each branch uses a short crossfade to reduce ON / OFF transients. Switching a branch off neither moves the thresholds nor extends the other branch into a different interval. In LR / MS, each channel has its own switches.

Ratio LINK: opposite proportional changes

The small LINK above and between the two Ratios appears only in DUAL.



Up and Down retain their existing relative relationship. Enabling LINK does not make values jump.

Ratio LINK defaults to on at first use, then remembers your last choice. When reopening a saved project, the LINK state saved in that project takes priority.

Initial Down / Up	Adjust Down	Linked Up
1:1 / 1:1	1:1 → 2:1	1:2
4:1 / 1:2	4:1 → 8:1	1:4
8:1 / 1:4	8:1 → 4:1	1:2

Preserve the relationship, without re-pairing

For example, doubling Down halves the Up value; adjusting Up uses the same inverse relationship. Starting at 4:1 and 1:2 will not force them into a reciprocal pair. Dragging, fine adjustment and direct entry all follow this rule.

At a limit, both stop together

Up ranges from 1:1000 to 1:1; Down from 1:1 to 1000:1. If either would exceed its range, both stop at the shared limit. Reverse the adjustment to continue. Toggling LINK alone does not change the Ratios.

To hear upward processing alone

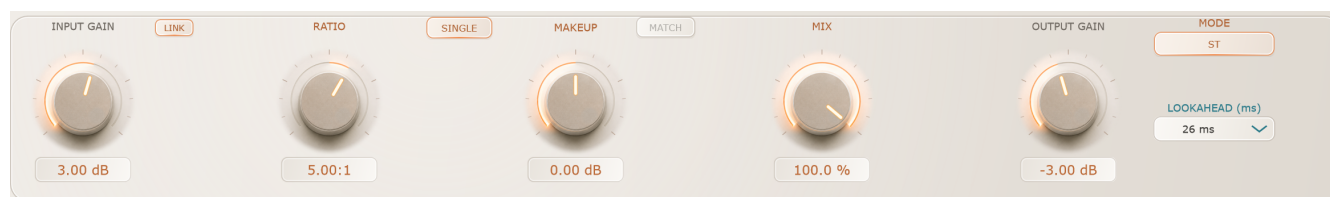
Switch DOWN off to preserve its Ratio while auditioning Up. To adjust Up Ratio independently as well, turn off the small Ratio LINK; otherwise the Down value still follows your adjustment.

Also available in LR / MS

The small LINK scales down with compact layouts and stays between Up / Down Ratio. It links those Ratios within each channel. Channel LINK beside Mode links L/R or M/S; see page 20.

Input / Output LINK: opposite gain changes

Above and to the right of INPUT GAIN. It is separate from Ratio LINK.



Example: Input +3 dB, Output -3 dB, with LINK enabled.

With LINK on, raising Input Gain in the interface lowers Output Gain by the same number of dB. Adjusting Output moves Input in the opposite direction too. This helps reduce simple level changes while you explore a different level entering the processor.

Initial Input / Output	Adjustment	Result
0 / 0 dB	Raise Input by 3 dB	+3 / -3 dB
+2 / -1 dB	Raise Input by 3 dB	+5 / -4 dB
+5 / -4 dB	Raise Output by 2 dB	+3 / -2 dB

Keep an existing offset

The link follows relative changes; it does not force the two values to be exact opposites. Toggling LINK alone leaves the gains unchanged. If either control reaches its allowed limit, both stop together.

Remember the last choice

LINK is on at first use. New instances then use your last choice, while saved projects restore their own state. This is a workflow preference and does not switch with A/B sound snapshots.

LINK is not Match

Input can change detection and compression; Output only trims the final level. Equal and opposite gain changes therefore do not guarantee equal loudness. Use Match or manual compensation when comparing dynamics.

Match and Makeup: compare at equal loudness

Control	Use
Makeup	Compensates Wet level before Mix; range -120 to +120 dB.
Output Gain	Trims the final mixed output; range -24 to +24 dB.
Match	Uses played material for a one-time loudness adjustment to Makeup. It is not continuous auto gain.

1 Play a representative passage

Allow a valid loudness reading to build before clicking Match. Use material that represents the passage, not just a very brief attack.

2 Compare Dry and Wet

Match compares Dry with Wet before Makeup. Makeup affects Wet; below 100% Mix, recheck the final blend by ear.

3 Trim the final level

Use Makeup or Output Gain for any small correction. Compare with Bypass or A/B and listen to attacks, tails and consistency across the passage.

Deep compression and very low levels

Deeply compressed Wet is no longer excluded just because it falls below a fixed loudness gate. The wider Makeup range also allows larger corrections. Match still needs a valid, nonzero measurement: silence or insufficient material cannot provide a reliable result.

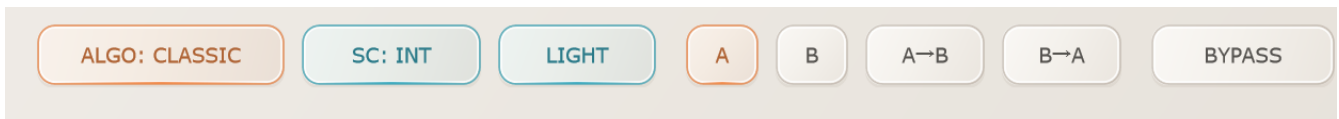
Recheck output after a large correction

Match is a comparison tool, not peak protection. Check the final level after deep compression, upward boost or further parameter changes so the next stage does not overload.

Three distinctions to keep in mind

Dynamic gain excludes Makeup and Output Gain. Input / Output LINK only makes equal and opposite gain changes; it cannot replace loudness matching. At Mix = 0%, Input and Output Gain still apply, so this is not full Bypass.

A/B: compare algorithms and directions



A and B hold two sound settings. A→B and B→A copy between them.

- 1 Save the first idea**
 Set the algorithm and processing amount in A. Copy A→B, then switch to B. Copying replaces the destination sound settings.
- 2 Change the part you want to compare**
 For example, use Classic in A and Super in B, or compare downward and upward processing. Keep Lookahead, detector source and monitoring the same where possible.
- 3 Match each, then switch**
 Play the same material for each setting and use Match; trim Makeup if needed. Click A / B and compare the sound, without mistaking louder for better.

Recalled by A/B

Algorithm, Single / Dual, thresholds, Ratios, Dual branch switches, gain, Mix, Lookahead, sidechain Source / Key Gain / HPF, and other sound settings.

Outside the A/B sound snapshot

Theme, LINK controls, Bypass and SC LISTEN.

Why can reciprocal Ratios sound the same?

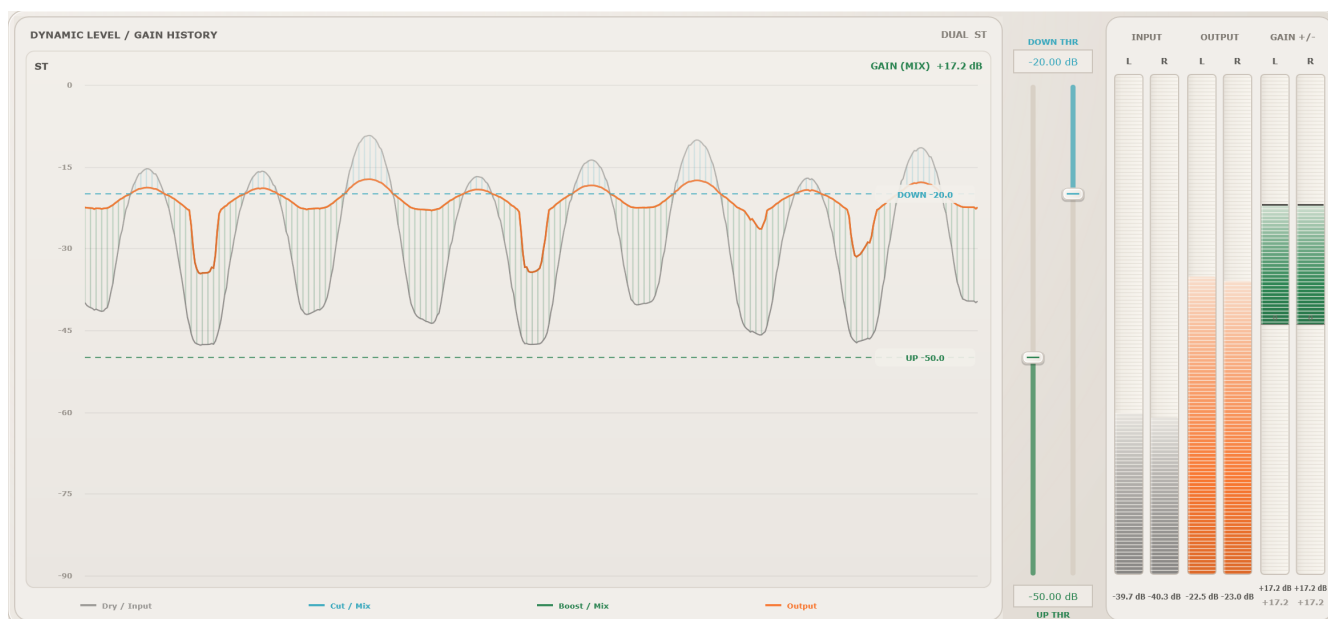
The upward and downward curves retain a symmetrical relationship. In a shared active interval, some reciprocal settings differ only by a constant gain, so they can sound very close after loudness matching. One example is 8:1 versus 1:8 in Super / SINGLE with Threshold = -inf, Range = OFF and Mix = 100%.

Classic also has a corresponding relationship within a shared active interval, away from the lower gate transition. Crossing a threshold, entering a finite Range transition, changing Mix or changing the Dual handoff can break equality across a whole passage.

Switching and latency

A/B uses transitions to reduce switching transients. Keep Lookahead the same for a clearer sound comparison. Different Lookahead settings may also cause the host to readjust delay compensation.

Read the Display and meters



In the same history view, green shows boost and Cut / Mix shows attenuation. Orange is the output-level reference.

What you see	Meaning
Dry / Input	Original dynamics before Input Gain; the grey line does not move as a whole when Input changes.
Cut / Mix	Dynamic attenuation including Mix. Its distance from Dry indicates the effective cut.
Boost / Mix	Dynamic boost including Mix, shown in green. Its distance from Dry indicates the amount of boost.
Output	Projected output-level reference under current settings, including Makeup, Mix and Output Gain.
External key	When EXT is selected and the external input is available, a lighter trace shows the sidechain reference to help check triggering.
Gain +/-	Zero is at the center: boost above, cut below. Positive values indicate boost and negative values indicate attenuation. Hold retains recent prominent changes.

Dynamic gain excludes Makeup / Output Gain and is 0 dB at Mix = 0%. All three themes show boost and cut; match the colors to the named legend below the graph.

History responds to current settings

Changing Ratio, thresholds, Range or Mix updates the visible history. EXT Key Gain updates immediately; HPF refreshes after you release it and may show HPF UPDATING. The graph recalculates historical level references under current settings. It is not a sample-by-sample recording of the output, nor does it replay audio.

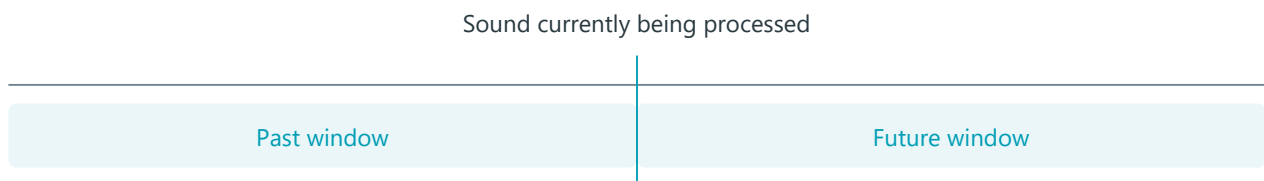
Lookahead: align gain with the sound

Classic and Super share the same detector timing.

Lookahead lets the plugin see a short section of upcoming signal before it outputs the current sound. A more stable level estimate helps reduce harmonics caused by gain following every sine-wave cycle. Lookahead is not conventional Attack.

How 1.2.6 reduces dips before a loud event

Detection considers windows before and after the current sound, then uses the lower of their two peaks. A future loud event no longer determines the attenuation of the preceding quiet passage on its own. When the loud event arrives, gain follows the aligned level estimate.



Schematic: both windows include the present; output is still delayed by Lookahead.

The default remains 26 ms

This detector change does not lengthen the default Lookahead, add an Attack / Release stage or add extra latency. Other window lengths remain available to suit the material.

Low distortion, with a trade-off on changing signals

A suitable window can give steady sine waves stable gain and preserve their shape. Music is not a constant sine wave: rapid amplitude modulation, short events and complex combinations can still produce modulation residuals. Lookahead does not guarantee zero harmonic distortion for every signal.

This version prioritizes less unnecessary advance attenuation so processing follows the current sound more closely. If a passage still shows small dips or waveform changes, check the detector source, window length and actual audio. Do not judge the result from a Display trace alone.

Lookahead and the 0 ms option

Lookahead affects level estimation and latency, not Attack. Longer windows may improve stability for some low-frequency or rapidly changing material, but do not improve every transient.

A longer window also increases plugin latency, which the host will usually compensate.



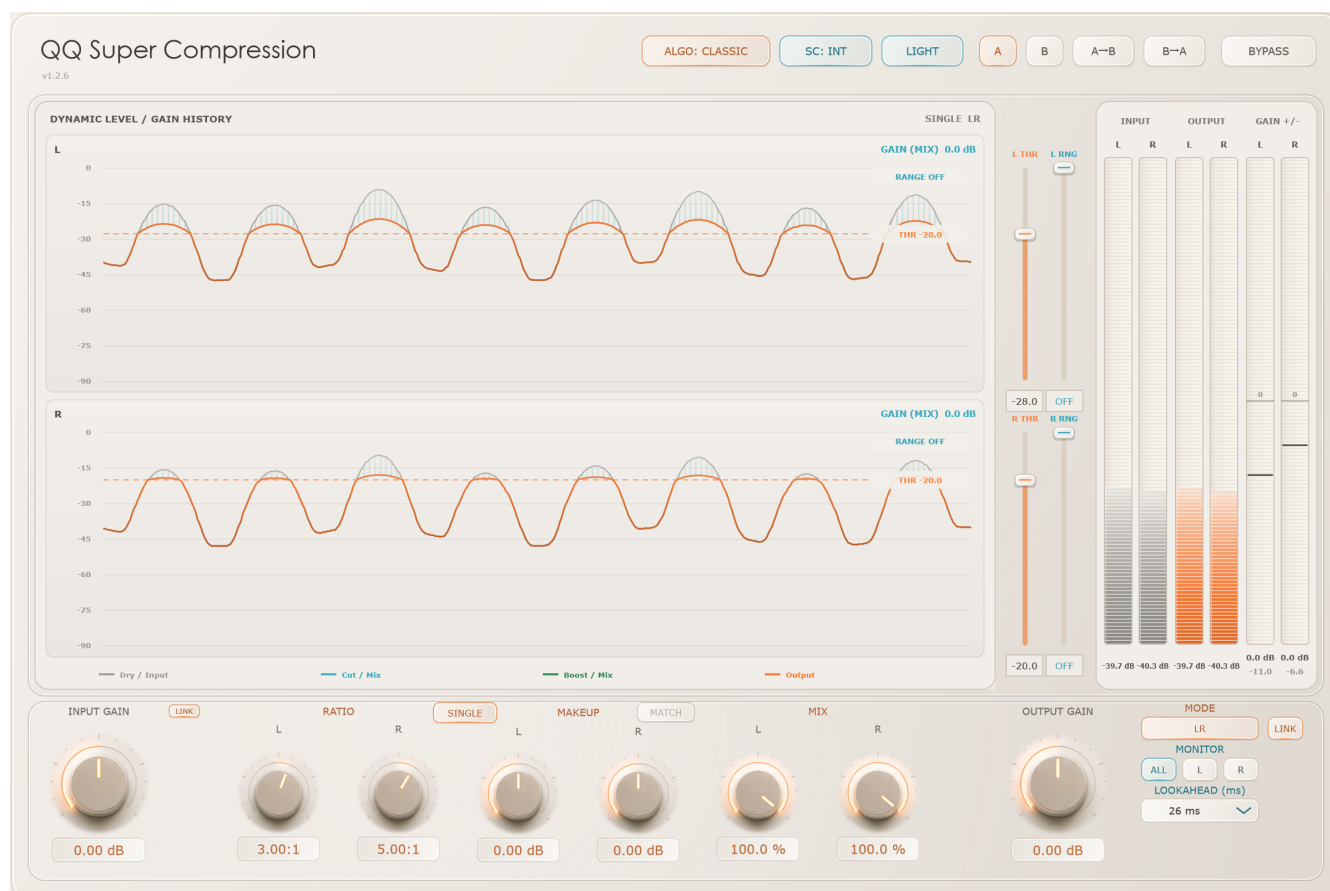
At 0 ms, Oversampling appears below Lookahead.

Choice	Use and trade-off
10 ms	Shorter window and lower latency.
26 / 40 ms	Default 26 ms. Compare with 40 ms if needed and choose by ear.
80 / 100 ms	Longer windows; listen for greater stability and consider the extra latency.
0 ms	More pronounced nonlinear colour, not the most transparent mode.

At 0 ms choose 1x / 8x / 16x Oversampling. Start with 8x; higher rates cost processing and latency. The control is hidden at 10 ms and above. 0 ms with 8x / 16x still adds latency.

ST and LR: together or independently

Click Mode at the lower right to cycle ST → MS → LR. Start with ST; choose LR when the two sides genuinely need different treatment.



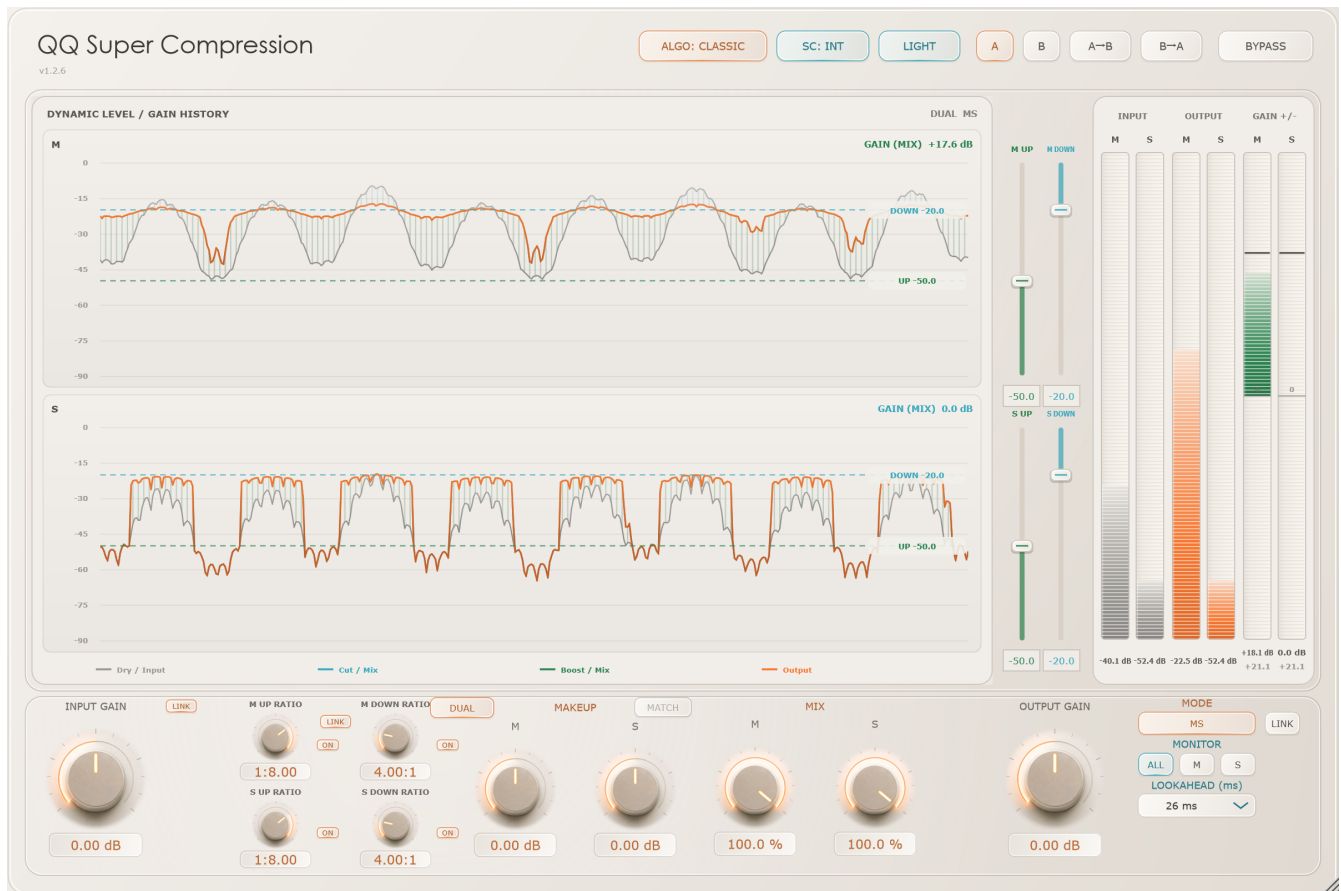
LR: L above, R below, with independent controls.

Mode	How to use it
ST	Uses common compression gain for stereo; a useful starting point for tracks and buses.
LR	Left and right have independent Ratio, boundary, Makeup and Mix controls. In DUAL, each side has its own UP / DOWN. Check for image shifts when processing asymmetric material.

MONITOR

Choose L or R to audition that side centred. Return to ALL to check normal stereo, and do not leave one-sided audition active for the final output.

MS: control centre and sides



MS / DUAL: Mid is above Side. Each channel has its own UP / DOWN Ratios and thresholds.

Mid mainly represents content common to both sides; Side represents their differences. They are not a precise separation of a centre track from every other track.

1 Listen to M and S first

Use M / S under MONITOR, then return to ALL.

2 Adjust each for a reason

For steadier centre dynamics, start with M Ratio / Mix. Treat S cautiously when preserving ambience and width.

Recheck the full mix

Compression or level changes in M/S can alter the sense of space. In ALL, check centre, width and mono compatibility, not only soloed domains.

Channel LINK: adjust together, keep differences

In LR or MS, LINK beside Mode links matching parameters across channels. Set the two sides separately, then enable it to adjust them together without forcing identical values.



SINGLE example: channel LINK beside Mode is on, while the left and right Ratios remain at 3:1 and 5:1.

Parameter	Before	After an equal change
Ratio	3:1 / 5:1	4:1 / 6:1
Threshold	-20 / -10 dB	-18 / -8 dB
Makeup	-3 / +1 dB	-2 / +2 dB
Mix	100% / 70%	90% / 60%

1 Set the relationship with LINK off

Decide which domain needs more compression or a different blend.

2 Enable LINK for joint adjustments

Move either parameter in a pair; the other follows by the same change. Fine adjustment and direct entry work too.

Both stop at a limit

If either side would exceed its range, both stop where the difference can be retained. In Super, one Threshold at $-\infty$ cannot form a finite dB offset; set both thresholds to finite values first if you want that relationship.

Three LINK controls, three roles

The small LINK between the Ratios links Up / Down within each channel. LINK beside Mode links matching L/R or M/S parameters. With both enabled, channels move together while each channel retains its existing Up / Down relationship. LINK beside Input only controls opposite input/output gain changes; see page 12.

Sidechain: start with the panel

Click SC: INT / SC: EXT at the top right. Its label shows the current source.



Control	How to use it
INT / EXT	INT follows the current track itself. EXT uses a signal sent from another track to control boost or attenuation on the current track.
KEY GAIN	Available only with EXT. Adjusts the external detector level and trigger strength, not the main track level.
HPF	Reduces low frequencies in the INT or EXT detector key. OFF is full-band; active range is 20-500 Hz.
KEY LEVEL	Check that the detector receives a signal. With EXT showing N/A, first check that the host sidechain bus is enabled.
SC LISTEN	Temporarily audition the detector key to check routing and HPF. Switch it off before judging normal output.

Two common tasks

Low notes pull the whole part down: try INT + HPF, page 22.

Want bass to duck when the kick plays? Use EXT, page 23.

Internal sidechain: reduce bass dominance

Use this when kick or bass repeatedly pulls down the rest of a drum bus, mix bus or master. The goal is less low-frequency triggering, not less bass in the audible signal.

- 1 Start with INT and hear the problem**
Loop a passage with bass and other content together. Leave HPF OFF to establish a comparison.
- 2 Open the sidechain panel**
Click SC: INT and keep SOURCE on INT. A disabled Key Gain is normal here.
- 3 Raise HPF gradually**
Start at a low cutoff and listen for less whole-mix ducking on bass events. Make small changes.
- 4 Audition the key if needed**
Use SC LISTEN to hear which bass content HPF removes, then turn it off and return to the full sound.
- 5 Inspect history after release**
HPF history refreshes after you release the knob. If HPF UPDATING appears, wait for it to clear before comparing.
- 6 Rebalance the compression amount**
Filtering may reduce GR or change its shape. Revisit Ratio, Threshold or Mix as needed, then compare at similar loudness.

It is normal not to hear bass being cut

HPF filters only the key used to decide compression, not normal output. It is not equivalent to Key Gain: Key Gain scales the detector level; HPF changes frequency weighting. Too high a cutoff can make the compressor ignore bass that still needs control.

External sidechain: kick-controlled bass

Insert the plug-in on the bass - the track to turn down. Send the kick to its sidechain as the trigger. Do not reverse the tracks or confuse a normal audio send with a sidechain send.



- 1 Enable the host sidechain input**
 Enable sidechain input for this plug-in instance on the bass. Add and enable a kick send to that sidechain. Names and locations vary by host.
- 2 Select EXT in the plug-in**
 Open SC, choose EXT and play the kick. Check KEY LEVEL first; if it has no reading, fix routing before increasing Ratio.
- 3 Confirm the actual trigger**
 Briefly enable SC LISTEN: you should hear the kick, not the bass. Turn audition off afterward.
- 4 Set trigger level and reduction**
 For downward ducking, start with SINGLE, Ratio above 1:1 and Range OFF. Set detector strength with Key Gain or the send level, and set the lower boundary with Threshold. Leave HPF OFF initially.
- 5 Use Mix to decide how much room to make**
 Find an audible duck, then reduce Mix to retain the desired bass sustain. Listen to both tracks together, not only solo bass.
- 6 Decide whether HPF helps**
 Raising HPF reduces low-frequency kick triggering and may weaken the duck you want. Use it when you want more emphasis on the attack or less unwanted low-bass triggering.

There are no conventional Attack / Release controls. Ducking shape depends on the trigger, Lookahead and compression settings; it is not guaranteed to be smoother than every conventional sidechain.

Apply it in a real mix

Vocals, guitar, piano: consistency with articulation

Try SINGLE / ST / INT and 26 ms. Raise Ratio above 1:1, then use Mix to keep consonants, plucks or hammer attacks natural. To process only louder parts, raise Threshold and leave Range OFF. Use Match and compare in the mix.

Tails and quiet phrases: bring detail forward

Try Single upward processing to lift quieter words, pluck tails or reverb. Use Threshold to exclude unwanted very quiet sounds and Range to set the upper limit. To also control louder parts, switch to Dual and adjust Up and Down separately. Use the branch ON / OFF switches to audition each direction. Turn Ratio LINK off if you also want to adjust each Ratio independently.

Mix bus / mastering: a more cohesive balance

When you want a steadier, more cohesive mix without deliberately reshaping drum attacks through conventional Attack / Release, try it as an alternative to G Bus-style compression. Start gently, compare balance across sections and lower Mix when needed.

If every kick pulls the whole mix down, try INT + HPF. Do not raise the cutoff merely to make the GR number smaller. At similar loudness, check drum attack, low end, space and the song's larger dynamic movement.

Vocal-keyed accompaniment: make space for words

Insert the plugin on the backing bus, send the vocal to its external sidechain and choose EXT. Confirm the source with SC LISTEN. Select SINGLE, Ratio above 1:1 and Range OFF, then adjust Key Gain, Threshold and Mix so the backing steps back while the vocal plays.

If plosives trigger too much ducking, try raising sidechain HPF. If breaths or noise also make the accompaniment dip, check the sent signal and Threshold; HPF alone will not solve every unwanted trigger.

For every comparison

Turn SC LISTEN off, return MONITOR to ALL, and compare at similar loudness. Use an appropriate limiter when you need a strict final peak ceiling; this plug-in is not a brick-wall limiter.

Quick reference: algorithms and dynamics

Parameter	Range / choices	Notes
ALGO	Classic / Super	Fixed dB ratio / original QQ curve. Remembers the last choice; project and A/B states take priority.
SINGLE / DUAL	Single / Dual	Button beside Ratio; each mode retains its own parameters.
Single Ratio	1:1000 to 1000:1	Default 1:1; below 1 upward, above 1 downward.
Up Ratio	1:1000 to 1:1	Default 1:1; upward boost.
Down Ratio	1:1 to 1000:1	Default 1:1; downward compression.
UP / DOWN switches	ON / OFF	Both default ON. Independent audition; Ratio values retained.
Threshold / UP THR	Classic: -90 to 0 dB; Super: -inf to 0 dB	Lower gate; defaults to the selected algorithm's lowest position.
Range	OFF or a set upper boundary	Single defaults to OFF. Set range: Classic -90 to 0 dB; Super -inf to 0 dB.
DOWN THR	Classic: -90 to 0 dB; Super: -inf to 0 dB	Dual handoff threshold; default 0 dB.
Ratio LINK	OFF / ON	Initially ON; remembers the last choice. Project state takes priority.
Mode / channel LINK	ST / MS / LR; OFF / ON	LINK joins matching parameters in LR / MS.

Boundaries and defaults

Range stays at or above Threshold; DOWN stays at or above UP. Faders push each other on contact; coincident boundaries stop dynamic processing. Existing projects keep their settings. All three Ratio types default to 1:1 in every channel.

Super's lowest finite threshold is -119.99 dB, followed by -inf at the bottom. The Display plots down to -90 dB. Range OFF is different from a finite 0 dB setting.

Quick reference: gain, monitoring and SC

Parameter	Range / choices	Notes
Input Gain	-24 to +24 dB	Level into the processing path.
Makeup	-120 to +120 dB	Default 0 dB; Wet level before Mix.
Mix	0 to 100%	Default 100%; included in effective dynamic gain.
Output Gain	-24 to +24 dB	Default 0 dB; final output level.
Input / Output LINK	OFF / ON	Initially ON. Equal opposite changes; remembers the last choice, with project state taking priority.
MONITOR	ALL / L / R or M / S	LR / MS audition; individual domain centred.
Lookahead	0 / 10 / 26 / 40 / 80 / 100 ms	Initial starting point 26 ms; new instances remember the last choice.
Oversampling	1x / 8x / 16x	Visible only at 0 ms; initially 8x.
SOURCE	INT / EXT	Default INT; EXT needs host routing.
KEY GAIN	-24 to +24 dB	Default 0 dB; adjustable only with EXT.
HPF	OFF / 20 to 500 Hz	Default OFF; filters only the detector key.
SC LISTEN	OFF / ON	Temporary audition; closing panel/editor switches it off.

Keep the gain controls distinct

Input changes the main input; Key Gain changes only the external detector; Makeup changes Wet before Mix; Output Gain changes the final output. Match loudness before comparing dynamics.

Upward / Dual troubleshooting

Symptom	First check
No audible boost after adjustment	Check that the relevant Ratio is below 1:1, Mix is above 0% and Bypass is off. In Dual, that branch must be ON.
No upward effect in Dual	The detector must be above UP and below DOWN. Below UP, the original level is retained; above DOWN, only downward processing applies.
Faders set, but no processing	Do the boundaries coincide? This stops all dynamic processing. Separate them before comparing.
Loud parts suddenly stop being processed	Is a finite Range set in SINGLE? At or above it, dynamic gain returns to 0 dB. Select OFF if you want no upper cutoff.
Adjusting Up also changes Down	LINK between the two Ratios is on by default. Turn it off to adjust them independently.
LINK will not move further	The other Ratio may have reached its limit, stopping both controls. Reverse the adjustment to continue.
Main track level does not explain triggering	Check whether SOURCE is EXT or HPF is changing the detector signal. Threshold decisions follow the detector signal.

Check Dual upward processing alone

- 1 Keep other variables fixed**
 Select DUAL, switch DOWN off and turn Ratio LINK off. Keep Mix at 100% and all gain controls at 0 dB. Choose a suitable Lookahead.
- 2 Place the signal between the thresholds**
 For example, set UP to -50 dB and DOWN to -20 dB. Check that the detector level of a quiet passage lies mainly between them. With Ratio still at 1:1, there is no boost yet.
- 3 Increase upward strength from 1:1**
 Move Up Ratio through 1:2, 1:4 and 1:8 while listening and watching Boost / Mix and positive dynamic gain. Once confirmed, enable DOWN to add attenuation and compare overall loudness again.

Troubleshooting

Why can Classic output fall below the threshold?

Threshold defines the gain curve for the detector level, not a minimum output level. Quieter sounds within a window can receive the gain set by its peak. Shared ST gain, external sidechain or HPF can also make the detector differ from the signal you see. Makeup and Output Gain then change the output further.

With Range OFF, internal detection, all gains at 0 dB and Mix at 100%, a steady level above threshold stays at or above it under the Classic formula. If a dip appears only in the Display, remember that it is a historical projection and check the actual audio. Not every dip is advance compression.

Symptom	First check
EXT has no effect	Enable the host sidechain input and send. Select EXT and check KEY LEVEL. N/A means no external bus is available.
Key Gain is disabled	Normal with INT: it adjusts only the external key.
HPF does not cut output bass	It filters the detector only. History refreshes after release; wait for HPF UPDATING to clear.
Sidechain sound / narrow image	Turn off SC LISTEN, return MONITOR to ALL, then check LR / MS.
Match is unavailable	Play enough non-silent material for valid readings. Falling below the old gate no longer disables matching.
Pause after changing Lookahead	Latency changes, so the host may readjust compensation. Also consider latency when monitoring a live recording.
Nothing below -90 dB	This is the Display range, not an audio cutoff.

For installation, updates and scanning, see the supplied Windows / macOS installation guide.

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github.com/Ziqing-Gu/QQ-Super-Compression